

RuSirius: An R Interface between the SIRIUS Mass Spectrometry Annotation Framework and The RforMassSpectrometry Ecosystem

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Summary

RuSirius is an R package that provides a programmatic interface between the SIRIUS mass spectrometry annotation framework ([Dührkop et al., 2019](#)) and the RforMassSpectrometry software ecosystem. SIRIUS is a widely used tool suite for the computational annotation of tandem mass spectrometry (MS/MS) data, encompassing molecular formula identification, compound class prediction, structure database search, *de novo* structure generation, and spectral library matching. The RforMassSpectrometry software ecosystem ([Rainer et al., 2022](#)) provides the main infrastructure for handling and analyzing MS data in R/Bioconductor. *RuSirius* connects to SIRIUS (version 6.4) through its REST API via the [RSirius client library](#), enabling users to control the full SIRIUS annotation pipeline from within R. By directly working on common R MS objects and converting them to SIRIUS-specific data structures, the package allows users to build end-to-end metabolomics workflows entirely within R.

Statement of Need

Untargeted metabolomics experiments generate large volumes of liquid chromatography–tandem mass spectrometry (LC-MS/MS) data that require computational annotation methods to assign putative identities to detected features. SIRIUS ([Dührkop et al., 2019](#)) is one of the most complete open-source frameworks for this task, integrating several distinct annotation tools. While SIRIUS provides a graphical user interface (GUI) and a command-line interface in Java, neither integrates well into scripted, reproducible data analysis workflows common in the R ecosystem. The R programming language is widely used in metabolomics, with established packages for data preprocessing (*xcms* ([Louail et al., 2025](#))), a powerful infrastructure for efficient and scalable MS data handling (*Spectra* ([Rainer et al., 2026](#)), *MsExperiment*), and annotation (*MetaboAnnotation*, *CompoundDb*) all available as part of the RforMassSpectrometry initiative ([Rainer et al., 2022](#)) and Bioconductor ([Huber et al., 2015](#)). Comparable computational annotation methods are however not available in R.

RuSirius addresses this gap by letting users import MS data as *Spectra* objects, run the full SIRIUS annotation pipeline, and retrieve results as R data frames — all from within a scripted R session. The targeted *RuSirius* users are metabolomics researchers who already use *xcms*

and the RforMassSpectrometry ecosystem for data preprocessing and need to integrate SIRIUS annotations into reproducible, end-to-end workflows. Finally, this integration also avoids the need of development of similar, but redundant, methodology in R.

State of the Field

Several tools exist for computational annotation of LC-MS/MS data, but few offer direct R integration. The *metfrag* R package (Ruttkies et al., 2016) provides in silico fragmentation-based annotation but does not cover formula identification, compound classification, or de novo structure generation. GNPS (Wang et al., 2016) offers web-based spectral library matching and molecular networking, but interaction from R requires manual file handling. The *RMassBank* package (Stravs et al., 2013) focuses on building spectral libraries from reference standards rather than annotating unknowns. SIRIUS provides the most comprehensive set of annotation tools available which include SIRIUS for molecular formula identification via fragmentation tree computation (Böcker & Dührkop, 2016), ZODIAC for network-based formula re-ranking (Ludwig et al., 2020), CSI:FingerID for molecular fingerprint prediction and structure database search (Dührkop et al., 2015), CANOPUS for compound class prediction (Dührkop et al., 2021), and MSNovelist for de novo structure generation (Stravs et al., 2022), and spectral library matching.

More recently, the SIRIUS developers introduced *RSirius*, an R client library for the SIRIUS REST API built on R6 classes. *RSirius* provides comprehensive API coverage, but it does not integrate with common MS data structures in R such as the one defined by the RforMassSpectrometry initiative. This software ecosystem (anchored by *xcms* (Louail et al., 2025) and *Spectra* (Rainer et al., 2022; Rainer et al., 2026)) follows a modular design in which independent packages interoperate through common data structures such as the *Spectra* object, enabling researchers to freely combine preprocessing, annotation, and statistical analysis tools into customizable and reproducible workflows. Rather than duplicating the API bindings already provided by *RSirius*, *RuSirius* builds on top of it, adding the higher-level, idiomatic R interface and data-structure integration that are needed to connect SIRIUS with this ecosystem. Together with *SpectriPy* (Graeve et al., 2025), *RuSirius* is part of ongoing RforMassSpectrometry efforts to improve integration and interoperability between MS data analysis tools across programming languages and platforms.

Software Design

RuSirius is structured around a central S4 class, *Sirius*, which encapsulates the connection to a running SIRIUS instance and tracks project state. This design reflects a key architectural trade-off: rather than reimplementing SIRIUS algorithms in R, *RuSirius* delegates computation to the SIRIUS server via its REST API and focuses on providing idiomatic R input/output. This keeps the package lightweight and ensures that users always run the same validated algorithms as the SIRIUS GUI or CLI.

The main user-facing functions follow the typical annotation workflow:

1. **Sirius()** establishes a connection, optionally authenticating and opening a project.
2. **import()** converts *Spectra* objects into SIRIUS feature imports, handling MS level grouping, adduct normalization, and chunked upload for large datasets.
3. **run()** submits annotation jobs with configurable parameter objects (*formulaIdParam*, *zodiacParam*, *predictParam*, *structureDbSearchParam*, *deNovoStructureParam*, *spectraMatchingParam*) that map directly to SIRIUS tool configurations. Prerequisite dependencies between tools are resolved automatically (e.g., requesting structure search enables fingerprint prediction and formula identification).
4. **results()** and **summary()** retrieve annotation results as data frames, supporting formula candidates, structure candidates, compound classes, *de novo* structures, spectral matches,

90 and fragmentation trees.
91 5. **Database management** functions (`listDbs()`, `createDb()`, `removeDb()`) allow users to
92 create and query custom spectral and structure databases.

93 Throughout this workflow, *RuSirius* maintains a mapping between user-facing feature identifiers
94 and SIRIUS-internal feature IDs, allowing seamless round-tripping between the R session and
95 the SIRIUS project.

96 A Docker image based on the official Bioconductor image is provided, bundling SIRIUS, *RSirius*,
97 and *RuSirius* with an automatically started REST API server, enabling reproducible deployment
98 without manual installation.

99 Research Impact Statement

100 *RuSirius* has been developed as part of the RforMassSpectrometry initiative and is hosted
101 under the *RforMassSpectrometry* GitHub organization. The package was developed in close
102 collaboration with the current SIRIUS developers, who recognized the need for an interoperable
103 bridge between *RSirius* and the RforMassSpectrometry software ecosystem. As documented by
104 the issue tracker on the GitHub repository, end-user feedback has guided the implementation
105 throughout development. The package has been used internally for metabolomics annotation
106 workflows at Eurac Research and Fondazione Edmund Mach, and applied in a ring trial
107 analysis (Louail & Rainer, 2026). Its use within a complete end-to-end metabolomics workflow
108 is demonstrated in the *Metabonaut* website (Louail et al., 2026), which applies *RuSirius*
109 to real-world data analysis alongside *xcms* and other RforMassSpectrometry packages. Its
110 integration with *xcms* and *Spectra* positions it as a seamless annotation step for the thousands
111 of researchers already using these packages for LC-MS/MS data processing.

112 AI Usage Disclosure

113 All content was written, reviewed, edited, and validated by the human authors. The core
114 software design decisions, implementation, and scientific framing are entirely the work of the
115 authors. Claude was used for reducing text length, for ensuring proper English phrasing and
116 correcting typos.

117 Availability

118 *RuSirius* is available on GitHub at <https://github.com/RforMassSpectrometry/RuSirius> under
119 the Artistic-2.0 license. Documentation for all exported functions and five tutorials (package
120 vignettes) covering the main workflows are available at <https://rformassspectrometry.github.io/RuSirius>. The *Metabonaut* website (Louail et al., 2026) demonstrates its use within a
121 complete end-to-end workflow. A Docker image with SIRIUS and all dependencies pre-installed
122 is available on Docker Hub at *rformassspectrometry/rusirius*.
123

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127 Conflicts of Interest

128 M.F., M.L., and S.B. are co-founders of Bright Giant GmbH.

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